Hriday Shah

London, UK | shah.hriday03@gmail.com | +44 7534 767193 | linkedin.com/in/hriday-shah03

EDUCATION

Imperial College London *Aug 2026 – Present*

Vellore Institute of Technology *2021 – 2025*
B.Tech, Computer Science and Engineering (Data Analytics Specialization) | GPA: 9.4/10 | Top 6, Department Rank
CFA Level I Candidate

TECHNICAL SKILLS & CERTIFICATIONS

Languages & Databases: Python, SQL, SAS, R, Java, C++, MySQL, PostgreSQL, MongoDB
Quant / ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, NLP, LLMs, Time Series (ARIMA/SARIMA), Anomaly Detection
AML & Compliance: SAS AML, SAS Visual Investigator, Transaction Monitoring, SAR Filing, KYC, Case Management
Tools: Docker, Git, Flask, REST APIs, Agile
Certifications: IBM Data Science Professional · Wharton Finance & Quantitative Modeling · Yale Financial Markets (Honors) · ISB Trading Strategies in Emerging Markets

WORK EXPERIENCE

SAS AML Developer / Trainee — Nupeak Pvt Ltd, Mumbai *Nov 2025 – Present*
- Reduced transaction monitoring processing time from ~25 hrs to 7–12 hrs by optimizing SAS AML workflows with Python and SQL, improving compliance SLA adherence
- Tuned AML detection scenarios and alert logic, reducing false positive rate and improving alert precision across monitored customer segments
- Built automated SQL validation pipelines for high-volume transaction feeds, reducing data quality incidents
- Streamlined SAR filing workflow design in line with regulatory frameworks

Data Science & ML Intern — Awarathon, Mumbai *Sep 2023 – Dec 2023*
- Built ML models to improve predictive accuracy on financial datasets; applied NLP/LLMs for sentiment-driven investment insight
- Optimized data pipelines to improve processing speed for faster downstream decision-making

Founder & AI/ML Developer, W.A.L.K Project — VIT-AP Incubation Center *May 2024 – 2025*
- Led development of an AI-driven biomedical device for drop-foot rehabilitation, improving measured patient gait outcomes

R&D Intern — Dubsync.ai, Hyderabad *Apr 2023 – Jul 2023*
- Engineered deep learning models for audio/video processing, improving lip-sync and generation accuracy

QUANTITATIVE PROJECTS

High-Frequency Implied Volatility Forecasting — Self Project *2025*
- Engineered 25+ microstructural order-book features to predict short-horizon implied volatility, outperforming baseline models
- Optimized a LightGBM pipeline for low-latency inference suitable for real-time signal generation
- Backtested volatility-based trading rules with cross-asset hedging to reduce simulated portfolio drawdown

Optimal Order Allocation & Order Flow Imbalance in Equity Markets — Self Project *2024 – 2025*
- Formulated a convex optimization model for order allocation across exchanges, reducing simulated execution costs
- Applied Robbins–Monro stochastic approximation to improve execution success vs. static baseline strategies
- Modeled order flow imbalance (OFI) via PCA to explain short-term price movements

Federated Learning for Data Privacy in Distributed Systems — VIT Research (Patent Filed) *2024*
- Developed a federated learning framework reducing centralized data dependency, improving cross-institutional data privacy
- Filed a patent application for secure, scalable ML deployment; co-authored under faculty guidance

Distributed Blockchain Consensus with Liquid Neural Networks — Patent Filed *2025*
- Designed a blockchain consensus mechanism using liquid neural networks for adaptive, real-time fraud detection in permissioned financial networks

ACHIEVEMENTS

- Hackathon Winner — 1st place, inter-college AI/ML solution hackathon
- Top 6, Department Rank — CSE (Data Analytics), VIT, Class of 2025
- Patents Filed (2) — Federated learning framework; blockchain–LNN consensus for financial compliance
- Research Publication — Co-authored faculty research on equity forecasting methodologies